2e. Content of Mechanics (Optional paper Y543)

Introduction to Mechanics.

In **Mechanics** learners extend their knowledge of particles, kinematics and forces from A Level Mathematics, using their extended pure mathematical knowledge to explore more complex physical systems. The area covers dimensional analysis, work, energy, power, impulse, momentum, centres of mass, circular motion and variable force.

6.01 Dimensional Analysis
The relationships between physical quantities are analysed by considering their dimensions (length, mass and time), in order to construct or check models.

6.02 Work, Energy and Power
The fundamental concepts of work, energy and power are introduced, including kinetic energy, gravitational potential energy and elastic potential energy. The principle of conservation of mechanical energy is used to solve problems.

6.03 Impulse and Momentum
Problems involving collisions in 1-D and 2-D are studied, using the principal of conservation of linear momentum and Newton's experimental law.

6.04 Centre of Mass
Rigid bodies are modelled as particles at their centres of mass. Techniques for finding the centre of mass of a body or system of bodies are explored, including integration.

6.05 Motion in a circle
The motion of a particle in a horizontal or vertical circle is explored, including motion which is not restricted to a circular path.

6.06 Further Dynamics and Kinematics
The techniques of differential equations studied in Pure Core are extended to linear motion under a variable force.

Assumed knowledge
Learners are assumed to know the content of GCSE (9–1) Mathematics and A Level Mathematics. They are also assumed to know the content of Pure Core (Y540 and Y541). All of this content is assumed, but will only be explicitly assessed where it appears in this section.

Resolving forces

The technique of **resolving forces** is found in 'Stage 2' of the A Level mathematics content, and therefore 'Stage 1' learners may not have learned this technique yet. As it is a vital underlying skill in the more advanced mechanics topics met in this paper, it is taken as assumed knowledge for 'Stage 1'. This includes both being able to express a force as two mutually perpendicular components, and being able to find the resultant of two or more forces acting at a point. See sections 6.02b, 6.02l and 6.05c.

Use of technology
To support the teaching and learning of mathematics using technology, we suggest that the following activities are carried out through the course:

1. Spreadsheets: Learners should use spreadsheets to generate tables of values for functions and to investigate functions numerically.

2. Learners should use graphing software for modelling, including kinematics and projectiles, and in visualising physical systems.

3. Computer Algebra System (CAS): Learners could use CAS software to investigate algebraic relationships, including derivatives and integrals, and as an investigative problem solving tool. This is best done in conjunction with other software such as graphing tools and spreadsheets.

4. Complex problem solving: Learners could use CAS to perform computation when solving complex problems in mechanics, including those which lead to equations or systems that they cannot solve analytically.

5. Practical mechanics: Learners could use computers and/or mobile phones to enrich practical mechanics tasks, using them for data logging, to create videos of moving objects, or to share and analyse data.

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When this course is being co-taught with AS Level Further Mathematics A (H235) the 'Stage 1' column indicates the common content between the two specifications and the 'Stage 2' column indicates content which is particular to this specification. In each section the OCR reference lists 'Stage 1' statements before 'Stage 2' statements.

OCR Ref.	Subject Content	Stage 1 learners should…	Stage 2 learners should additionally…
6.01 Dimensional Analysis			
6.01a	Dimensional analysis	a) Be able to find the dimensions of a quantity in terms of M, L and T, and understand that some quantities are dimensionless. *Includes understanding and using the notation $[d]$ for the dimension of the quantity d.* *Learners are expected to know or be able to derive the dimensions of any quantity for which they know the units. Dimensions of other quantities will be given, or their derivation will be the focus of assessment.*	
6.01b		b) Understand and be able to use the relationship between the units of a quantity and its dimensions.	
6.01c		c) Be able to use dimensional analysis as an error check. *e.g. Verify the relationship that power is proportional to the product of the driving force and the velocity.*	
6.01d		d) Be able to use dimensional analysis to determine unknown indices in a proposed formulation. *e.g. Determine the period of oscillation of a simple pendulum in terms of its length, mass and the acceleration due to gravity, g.*	
6.01e		e) Be able to formulate models and derive equations of motion using a dimensional argument.	

OCR Ref.	Subject Content	Stage 1 learners should…	Stage 2 learners should additionally…